



ET 8311

TELE TRAFFIC ENGINEERING

Name: Joseph Sospeter Salawa

Contacts: 0754556575

Email: yusufujss@gmail.com

Introduction

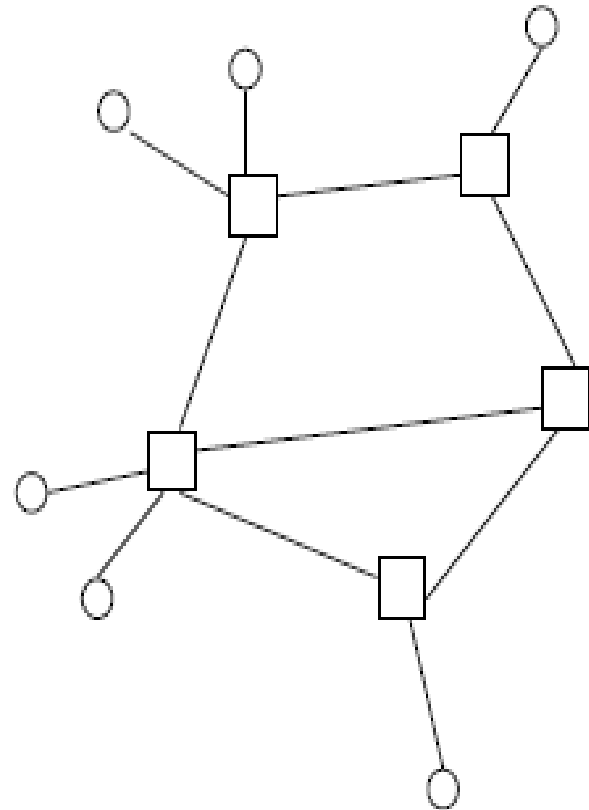
- Teletraffic theory is defined as *the application of probability theory to the solution of problems concerning planning, performance evaluation, operation and maintenance of telecommunication systems.*
- More generally, teletraffic theory can be viewed as a discipline of planning where the tools (stochastic processes, queuing theory and numerical simulation) are taken from operations research.
- The term *teletraffic* covers *all kinds of data communication traffic and telecommunication traffic.*

Cont...

- The objective of teletraffic theory can be formulated as follows:
- *to make the traffic measurable in well defined units through mathematical models*
- *derive the relationship between grade-of-service and system capacity in such a way that the theory becomes a tool by which investments can be planned".*

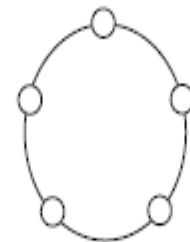
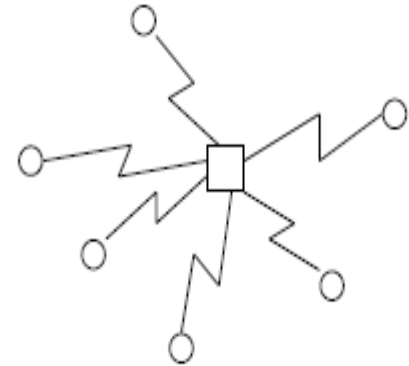
Telecom Networks

- A simple model of a telecommunication network consists of
 - nodes
 - terminals
 - network nodes
 - links between nodes
 - Access network
 - connects the terminals to the network nodes
 - Trunk network
 - connects the network nodes to each other



Cont...

- In the previous model,
 - connections between terminals and network nodes are point-to point type (no resource sharing within the access netw.)
- In some cases, such as
 - mobile telephone network
 - local area network (LAN) connecting computers the access network consists of shared medium:
 - users have to compete for the resources of this shared medium
 - multiple access (MA) techniques are needed

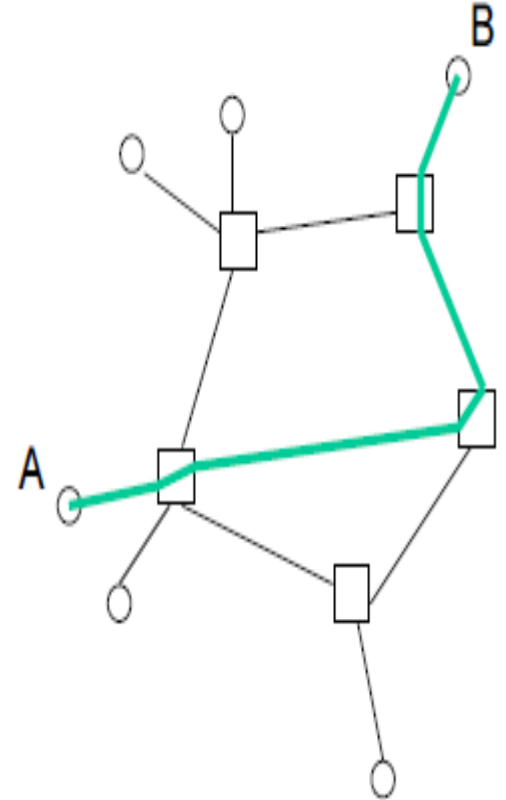


Switching modes

- Circuit switching
 - telephone networks
 - mobile telephone networks
 - optical networks
- Packet switching
 - data networks
 - two possibilities
 - connection oriented: e.g. X.25, Frame Relay
 - connectionless: e.g. Internet (IP), SS7 (MTP)
- Cell switching
 - ATM networks
 - connection oriented
 - fast packet switching with fixed length packets (cells)

Circuit switching

- Connection oriented:
 - connections set up end-to-end before information transfer
 - resources reserved for the whole duration of connection
 - if resources are not available, the call is blocked and lost
- Information transfer as continuous stream

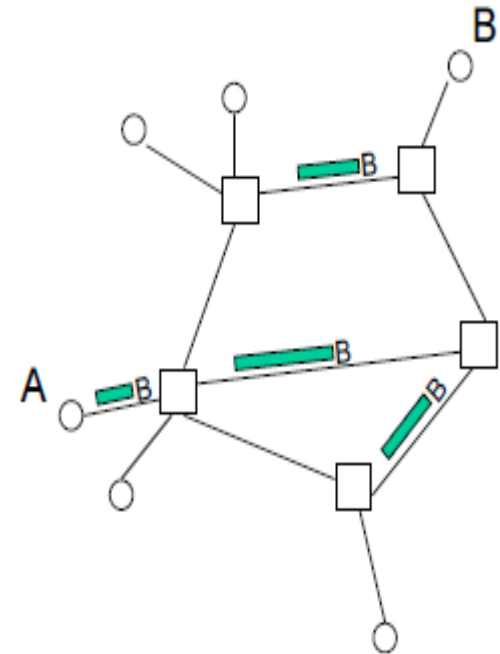


Cont...

- Before information transfer
 - Set-up delay
- During information transfer
 - signal propagation delay
 - no overhead
 - no extra delays
- Example: telephone network

Connectionless packet switching

- Connectionless:
 - no connection set-up
 - no resource reservation
 - no blocking
- Information transfer as discrete packets
 - varying length
 - global address (of the destination)



Cont...

- Before information transfer
 - no delays
- During information transfer
 - overhead (header bytes)
 - packet processing delays
 - queuing delays (since packets compete for joint resources)
 - transmission delays (due to finite capacity links)
 - signal propagation delay
 - packet losses (due to finite buffers)
- Example: Internet (IP-layer)

Traffic point of view

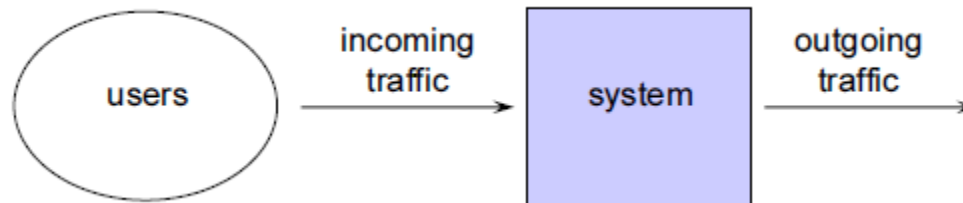
- Telecommunication system from the traffic point of view



- Ideas:
 - the system serves the incoming traffic
 - the traffic is generated by the users of the system

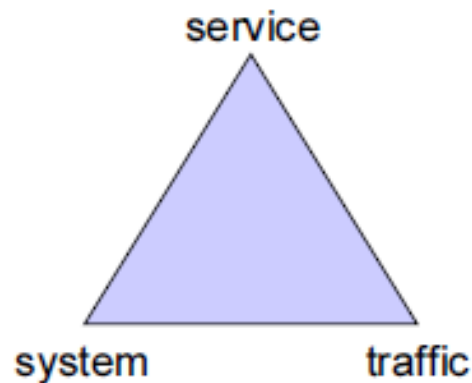
Interesting questions

- Given the system and incoming traffic, what is the quality of service experienced by the user?
- Given the incoming traffic and required quality of service, how should the system be dimensioned?
- Given the system and required quality of service, what is the maximum traffic load?



Cont...

- Determine relationships between the following three factors:
 - quality of service
 - traffic load
 - system capacity

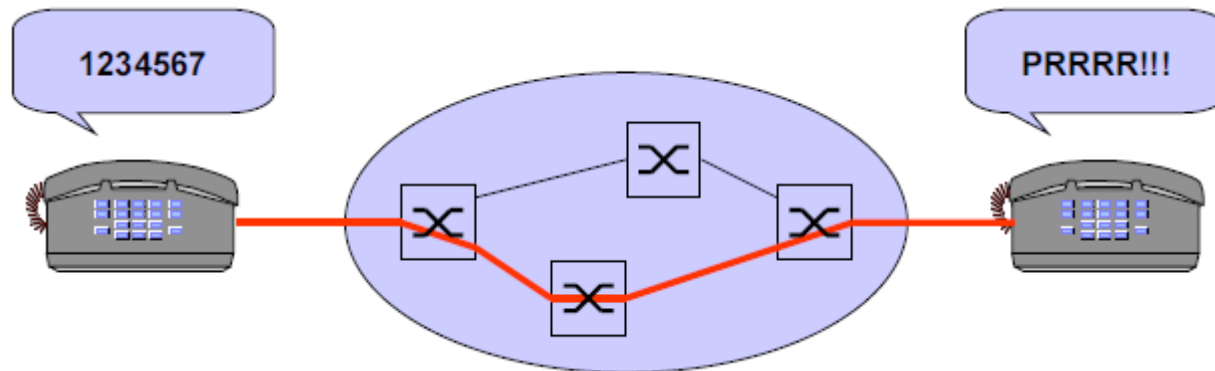


Cont...

- System can be
 - a single device (e.g. link between two telephone exchanges, link in an IP network, packet processor in a data network, router's transmission buffer, or statistical multiplexer in an ATM network)
 - the whole network (e.g. telephone or data network) or some part of it
- Traffic consists of
 - bits, packets, bursts, flows, connections, calls, ...
 - depending on the system and time scale considered
- Quality of service can be described from the point of view of
 - the customer (e.g. call blocking, packet loss, packet delay, or throughput)
 - the system, in which case we use the term performance (e.g. processor or link utilization, or maximum network load)

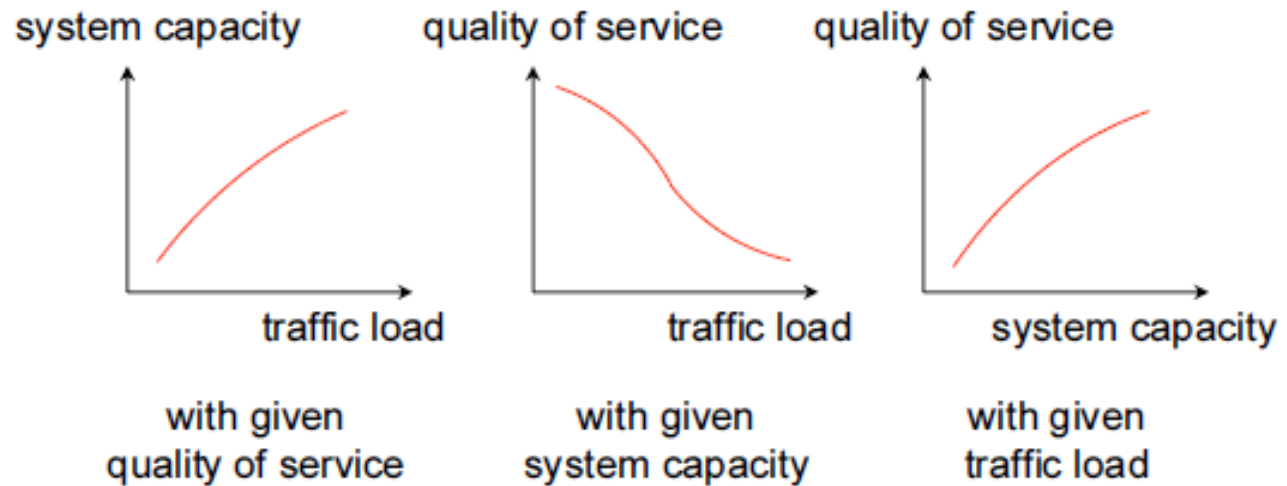
Example

- Telephone call
 - traffic = telephone calls by everybody
 - system = telephone network
 - quality of service = probability that the phone rings at the destination



Relationships between the three factors

- Qualitatively, the relationships are as follows:



- To describe the relationships quantitatively, mathematical models are needed

Teletraffic models

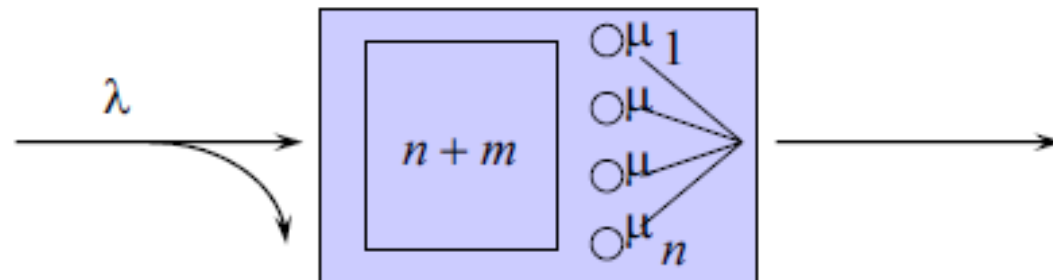
- Teletraffic models are stochastic (= probabilistic)
 - systems themselves are usually deterministic but traffic is typically stochastic
 - “you never know, who calls you and when”
- It follows that the variables in these models are random variables, e.g.
 - number of ongoing calls
 - number of packets in a buffer
- Random variable is described by its distribution, e.g.
 - probability that there are n ongoing calls
 - probability that there are n packets in the buffer
- Stochastic process describes the temporal development of a random variable.

Cont...

- Three types of system models:
 - loss systems
 - queuing systems
 - sharing systems
- These models can be combined to create models for whole telecommunication networks
 - loss networks
 - queuing networks
 - sharing networks

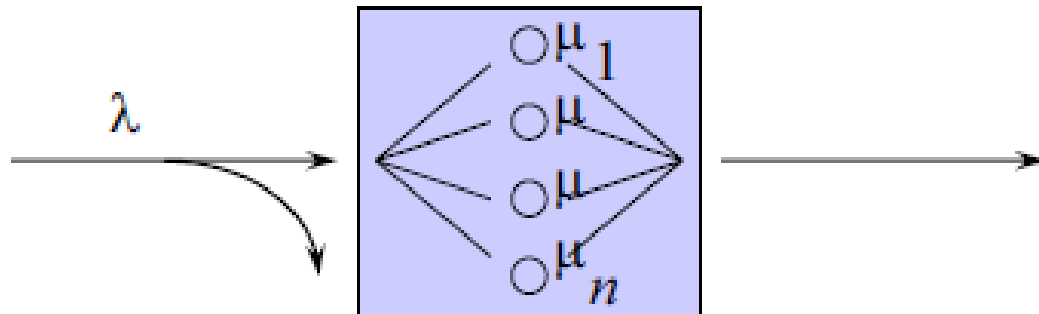
Simple teletraffic model

- **Customers arrive at rate λ** (customers per time unit)
 - $1/\lambda$ = average inter-arrival time
- Customers are **served by n parallel servers**
- When busy, a server serves at rate μ (customers per time unit)
 - $1/\mu$ = average service time of a customer
- There are **$n + m$ customer places** in the system
 - at least n service places and at most m waiting places
- It is assumed that blocked customers (arriving in a full system) are lost



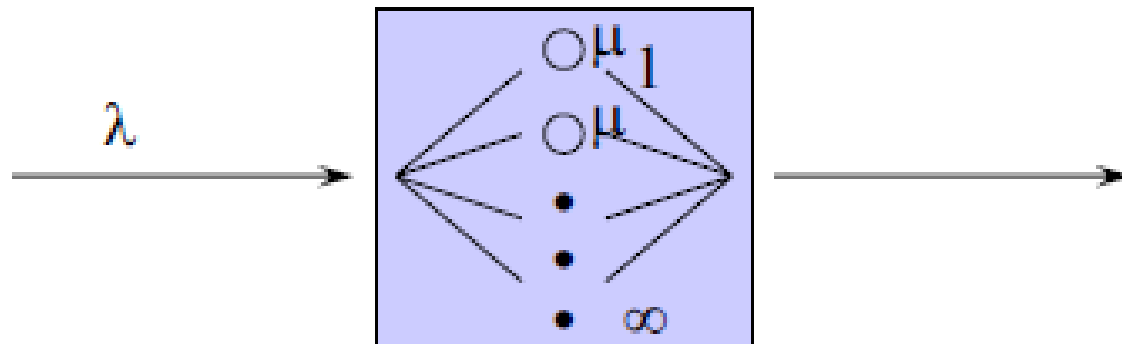
Pure loss system

- Finite number of servers ($n < \infty$), n service places, no waiting places ($m = 0$)
 - If the system is full (with all n servers occupied) when a customer arrives, it is not served at all but lost
 - Some customers may be lost
- From the customer's point of view, it is interesting to know e.g.
 - What is the probability that the system is full when it arrives?



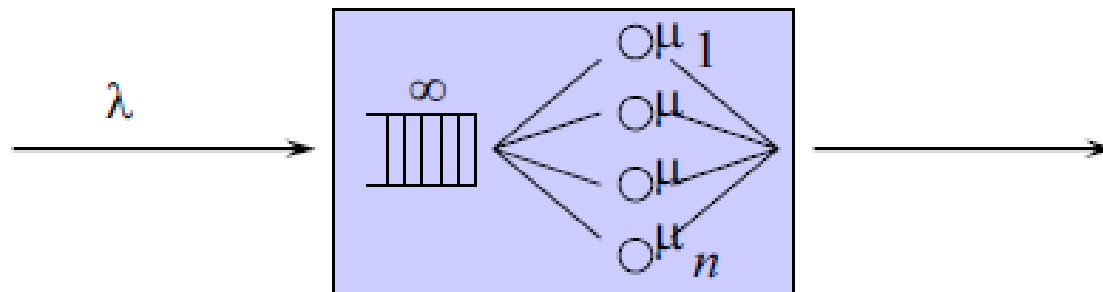
Infinite system

- Infinite number of servers ($n = \infty$), no waiting places ($m = 0$)
 - No customers are lost or even have to wait before getting served
- Sometimes,
 - this hypothetical model can be used to get some approximate results for a real system (with finite system capacity)
- Always,
 - it gives bounds for the performance of a real system (with finite system capacity)
 - it is much easier to analyze than the corresponding finite capacity models



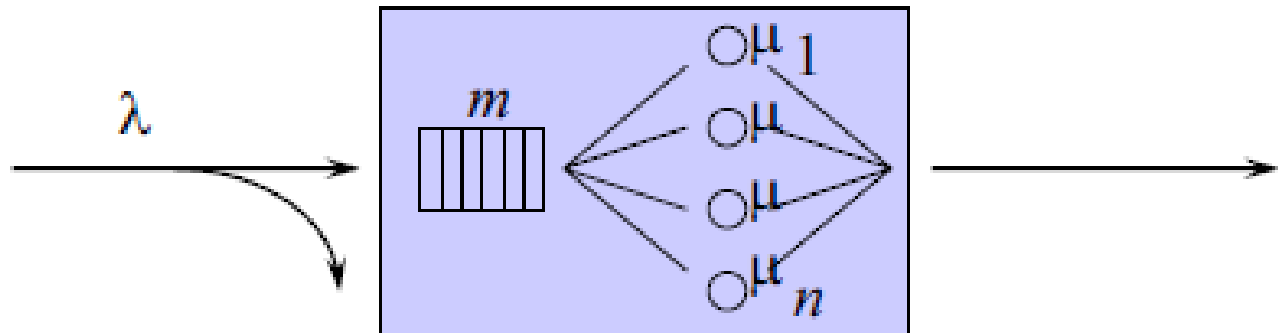
Pure queuing system

- Finite number of servers ($n < \infty$), n service places, infinite number of waiting places ($m = \infty$)
 - If all n servers are occupied when a customer arrives, it occupies one of the waiting places
 - No customers are lost but some of them have to wait before getting served
- From the customer's point of view, it is interesting to know e.g.
 - what is the probability that it has to wait "too long"?



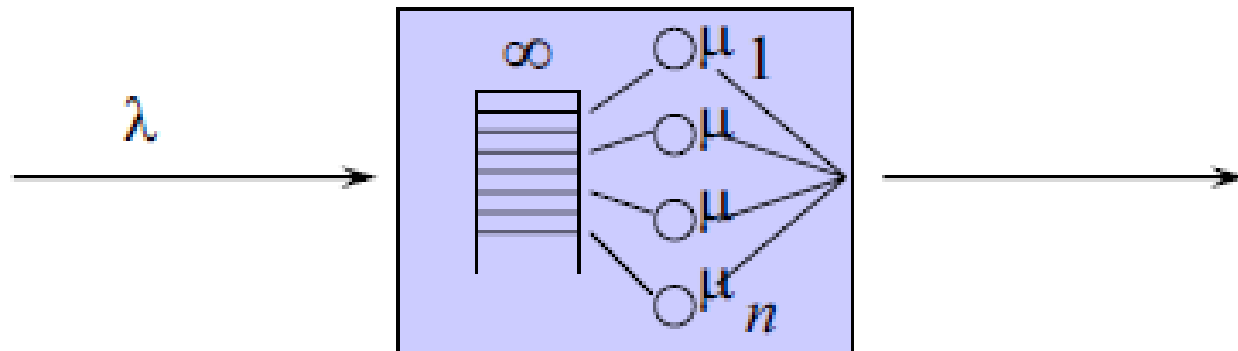
Lossy queuing system

- Finite number of servers ($n < \infty$), n service places, finite number of waiting places ($0 < m < \infty$)
 - If all n servers are occupied but there are free waiting places when a customer arrives, it occupies one of the waiting places
 - If all n servers and all m waiting places are occupied when a customer arrives, it is not served at all but lost
 - Some customers are lost and some customers have to wait before getting served



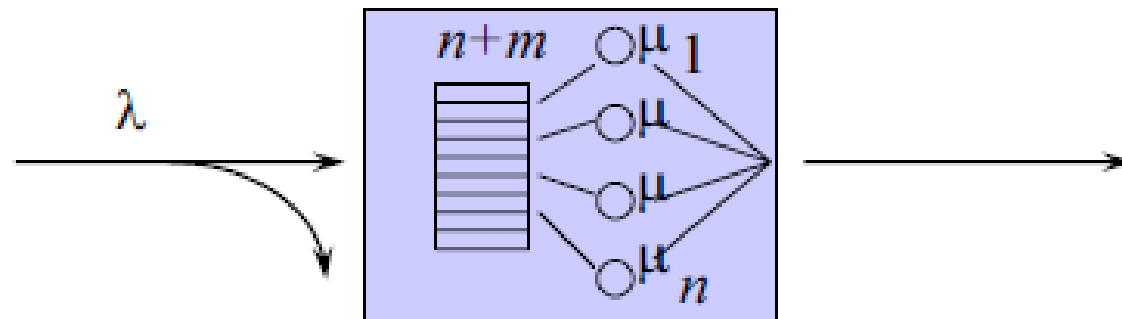
Pure sharing system

- Finite number of servers ($n < \infty$), infinite number of service places ($n + m = \infty$), no waiting places
 - If there are at most n customers in the system ($x \leq n$), each customer has its own server. Otherwise ($x > n$), the total service rate ($n\mu$) is shared fairly among all customers.
 - Thus, the rate at which a customer is served equals $\min\{\mu, n\mu/x\}$
 - No customers are lost, and no one needs to wait before the service.
 - But the delay is the greater, the more there are customers in the system. Thus, delay is an interesting measure from the customer's point of view.



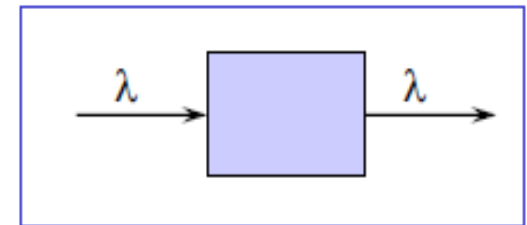
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 - Some customers are lost, but no one needs to wait before the service.



Little's formula

- Consider a system where
 - new customers arrive at rate λ
- Assume **stability**:
 - Every now and then, the system is empty
- Consequence:
 - Customers depart from the system at rate λ
- Let
 - $\bar{N}$ = average number of customers in the system
 - $\bar{T}$ = average time a customer spends in the system = average delay
- Little's formula:



$$\bar{N} = \lambda \bar{T}$$